A Study on the Compatibility of Gerrymandering Metrics

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***Abstract*— Gerrymandering has been a very controversial topic in politics and law in the past decade. There is no single universal metric to measure Partisan Gerrymandering. The four most widely used metrics-Efficiency Gap (EG), Declination, Partisan Bias, and Mean Median Difference (MMD)- each have their own characteristics. A strong legal definition of a Gerrymander would likely encompass several of them. In order to do this, the metrics used must be compatible. This paper examines how they relate to each other in the context of the 2020 redistricting maps in the USA by running Pearson correlation and linear regression tests. Results showed that Partisan Bias and Declination, and Partisan Bias and MMD were not compatible at all ($p = 0.656$, 0.198), while the rest were somewhat compatible ($p < 0.001$). Of these, Declination and MMD, and Declination and EG were most compatible ($R^2 = 0.37, 0.48$). Given the size of the dataset, the results encompass a small portion of scenarios, so running simulations over a wide range of scenarios for the maps would be an excellent way to build upon this work.**

Key Terms* — *Gerrymandering, Metrics, Efficiency Gap, Declination, Partisan Bias, Mean-Median Difference

I. Introduction

Gerrymandering is the means by which a redistricting map becomes biased toward one political party. This practice was first applied in 1812 by then Massachusetts Governor Elbridge Gerry, who was said to have manipulated redistricting to favor his party and thus gave one of the shapes of a district a salamander-like shape. Since this time, Gerrymandering has been a dyed-in-the-wool technique in American politics, often resulting in disproportional representation in legislative bodies. This has been made clear through recent court decisions, such as Covington v. North Carolina [1], Common Cause v. Lewis [2] and Gill v. Whitford [3], in which the contentious nature of Gerrymandering, and, more importantly, the legal definition to detect it, becomes noteworthy.

Some of the metrics that have been developed in detecting Partisan Gerrymandering include the Efficiency Gap (EG) [4], Declination [5], Partisan Bias [6], and Mean-Median Difference (MMD) [7].

Metric Definitions:

Definition 1. Efficiency Gap (EG)

For each district, consider all votes beyond 50% for the winning party as wasted for them, and all votes for a losing candidate as wasted for their party. Sum the wasted votes for each party over all districts in a state. Find the difference between these wasted votes for the two parties and divide this difference by the total number of votes. This yields the Efficiency Gap (EG) [4].

Definition 2. Declination

Sort the seats in ascending order of vote share for one of the parties. Plot the number of seats vs the party's vote share. Then take the respective centers of mass (R and D) of seats that each party won, as well as the point at 50% vote share between the two sets of data (Win/Loss). Take the angles of the lines between each center of mass and the Win/Loss point and find the difference in the angles (D-R. This yields the Declination angle. The angle can also be multiplied by 2 to keep its range from -1 to 1 [5].

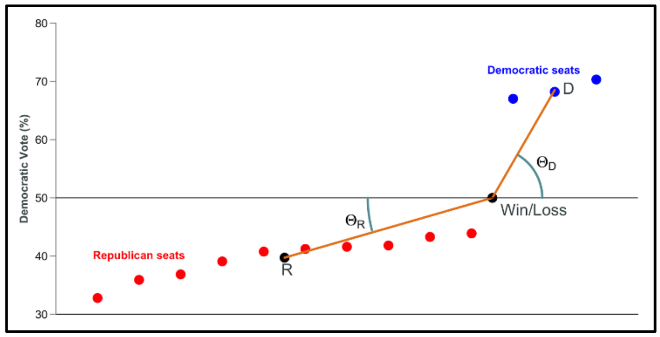

Figure 1. Geometric Explanation of Declination. *(figure caption)*

Definition 3. Partisan Bias

Take a state and apply a uniform shift in results to all districts until the statewide vote share is tied. Then find the percentage of seats each party gets in this hypothetical scenario and find the difference. This yields Partisan Bias [6].

Definition 4. Mean-Median Difference (MMD)

Sort the districts by their vote share in favor of one of the parties. Find the median district and find the difference between the party's vote share and 50%. This yields MMD [7].

These metrics sometimes return differing results, which raises the question of which ones should be used. All 4 of the metrics have been shown to be imperfect [8][9][10]. In order for a set of multiple metrics to give a legal definition, they firstly need to agree in most scenarios to ensure that it's reasonably possible to create a fair map. Secondly, they need to disagree occasionally, so that in cases where one metric fails, another one succeeds. This means that the metrics should all be correlated and need to be moderately or strongly correlated. This paper gives insights into their compatibility by making use of data from the redistricting maps of the 2020s.

II. Methods

This analysis compares the four Gerrymandering detection metrics of EG, Declination, Partisan Bias, and MMD, using data derived from official redistricting maps for the 2020s. The various metrics were computed based on the mean of the outcomes of the 2016 and 2020 presidential elections [11].

Pearson correlation tests for positive correlation were conducted to examine the relationship between pairs of metrics and to see if the correlation was statistically significant. Univariate Linear Regression Analysis for each of the 6 possible pairs was conducted to gauge the strength of these relationships.

Results

The correlations varied from fair to poor. Four of the pairs were moderately strongly positively correlated with one another. The 2 other metric pairs, Partisan Bias and MMD, and Partisan Bias and Declination were not positively correlated. The findings for the 6 pairs are presented in the tables below.

TABLE I. Pearson Correlation and R-Squared For All Metric Pairs (N = 111)

Metric Pair	*Pearson Correlation Coefficient (r)*	*p-value*	*R^2*
EG & Declination	0.69	<0.001***	0.48
EG & Partisan Bias	0.36	<0.001***	0.13
EG & MMD	0.35	<0.001***	0.12
Declination & Partisan Bias	-0.04	0.656	0.00
Declination & MMD	0.61	<0.001***	0.37
Partisan Bias & MMD	0.08	0.198	0.01

* = p < 0.05, ** = p < 0.01, *** = p < 0.001

The tables above show that Partisan Bias and Declination, and Partisan Bias and MMD are not positively correlated, while Partisan Bias and EG, and MMD and EG are relatively weakly correlated. Meanwhile, Declination and EG, and Declination and MMD are moderately strongly correlated.

In addition to the correlation analysis, regression analyses were conducted to further explore the strength of the relationships between the metrics. The regression results are summarized in the table below.

TABLE II. Linear Regression Analyses For All Metric Pairs (N = 111)

Source	*B*	*SE B*	*p-value*	R^2
Constant	8.43	1.16	<0.001***	
Declination vs EG	1.16	10.02	<0.001***	0.48
Constant	-0.12	0.52	0.814	
Partisan Bias vs EG	0.3	0.07	<0.001***	0.13
Constant	2.62	0.27	<0.001***	
MMD vs EG	0.15	0.04	<0.001***	0.12
Constant	0.55	0.66	0.406	
Partisan Bias vs Declination	-0.01	0.03	0.689	0.00
Constant	1.63	0.27	<0.001***	
MMD vs Declination	0.11	0.01	<0.001***	0.37
Constant	2.86	0.28	<0.001***	
MMD vs Partisan Bias	0.04	0.05	0.397	0.01

* = p < 0.05, ** = p < 0.01, *** = p < 0.001

Discussion

Efficiency Gap and Declination present the strongest relationship with an R^2 of 0.48, which suggests that these are very compatible, agreeing in most, but not all scenarios. It's a similar story for Declination and MMD, with an R^2 of 0.37. Conversely, Partisan Bias exhibits almost no positive correlation with Declination and MMD, as the p-values for the Pearson positive correlation tests were not statistically significant at 0.656 and 0.198, respectively. Efficiency Gap was positively correlated with Partisan Bias and MMD, but the low R^2 values of 0.13 and 0.12 suggest that the pairs are not very strongly compatible, as the relationship only accounts for approximately one eighth of variance.

These findings provide more grounds for a set of different metrics to be relevant in relation to the proper detection of Gerrymandering. While EG and Declination are recommended on the basis of greater reliability, the study also indicates a potential for Mean Median Difference to be applied cautiously, especially in the less competitive districts. Overall, the strongest pairs of metrics were Declination and EG and Declination and MMD. These results serve as proof of concept for a framework to assess metric compatibility.

Prior literature has found that the mathematical definition of Declination and EG are similar, as both are based on measuring the proportion of votes that are wasted in an election, supporting the strong result of Declination and EG [10]. The only other pair of metrics that have a mathematical connection are Partisan Bias and MMD, as they are both based on applying shifts and examine seat share with equal vote share and vote share with equal seat share respectively. This connection, however, is quite weak. The other 4 metrics pairs don't have mathematical similarities according to prior literature.

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Conclusion

Results from this study indicate that Declination and Efficiency Gap are the metrics most compatible with detecting Partisan Gerrymandering, while Declination and MMD were also strongly compatible. Other metrics, -Partisan Bias and mean-median difference-are weaker in correlation and therefore could present some limitations in their usability. It is important to note that the sample size of elections is quite small at 111, using only a few maps. The results here are generally effective at showing that two metrics are entirely incompatible. For the 4 pairs that were deemed somewhat compatible, especially the 2 strongest pairs (Declination and Efficiency Gap, and Declination and MMD), the results are by no means conclusive enough to suggest that these two are appropriate for a legal definition. Given the lack of any mathematical or statistical connection between Declination and MMD, the results could be a fluke due to small test size, but more research is needed to know. However, the main takeaways are that the pairs of Partisan Bias and Declination, and Partisan Bias and MMD are not compatible, giving strong evidence that Partisan Bias is not a suitable metric. Likewise, given that Declination was part of the 2 strongest metric pairs, there is some evidence, albeit not statistically significant, to suggest that Declination is the most suitable metric. There is room for the framework for compatibility to be expanded. One such example would be to run MonteCarlo simulations to see how the metrics compare over a wide range of scenarios, encompassing states of various partisan leans, various types of Gerrymanders and various projected deviations from real results, which would be very computationally intensive.

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